HR Analytics & Reporting Sub-System

Project Brief

71% of companies consider people analytics a high priority, and 31% rate it as very important. HR analytics quantifies key HR aspects such as employee turnover, compensation, engagement, headcount, and demographics. By analyzing this data, organizations can optimize workforce strategies to foster growth, productivity, and employee satisfaction.

The objective of this sub-system is to design and develop a minimal yet functional HR Analytics and Reporting module. This module processes raw HRMS data and transforms it into meaningful insights through dashboards, charts, tables, and reports. The system enables data-driven decision-making by providing visual analytics such as employee growth trends, attrition rates, and workforce metrics.

This document contains:
- Component Diagram of the HR Analytics Backend Sub-System
- UI Wireframe Diagram of the HR Analytics Dashboard

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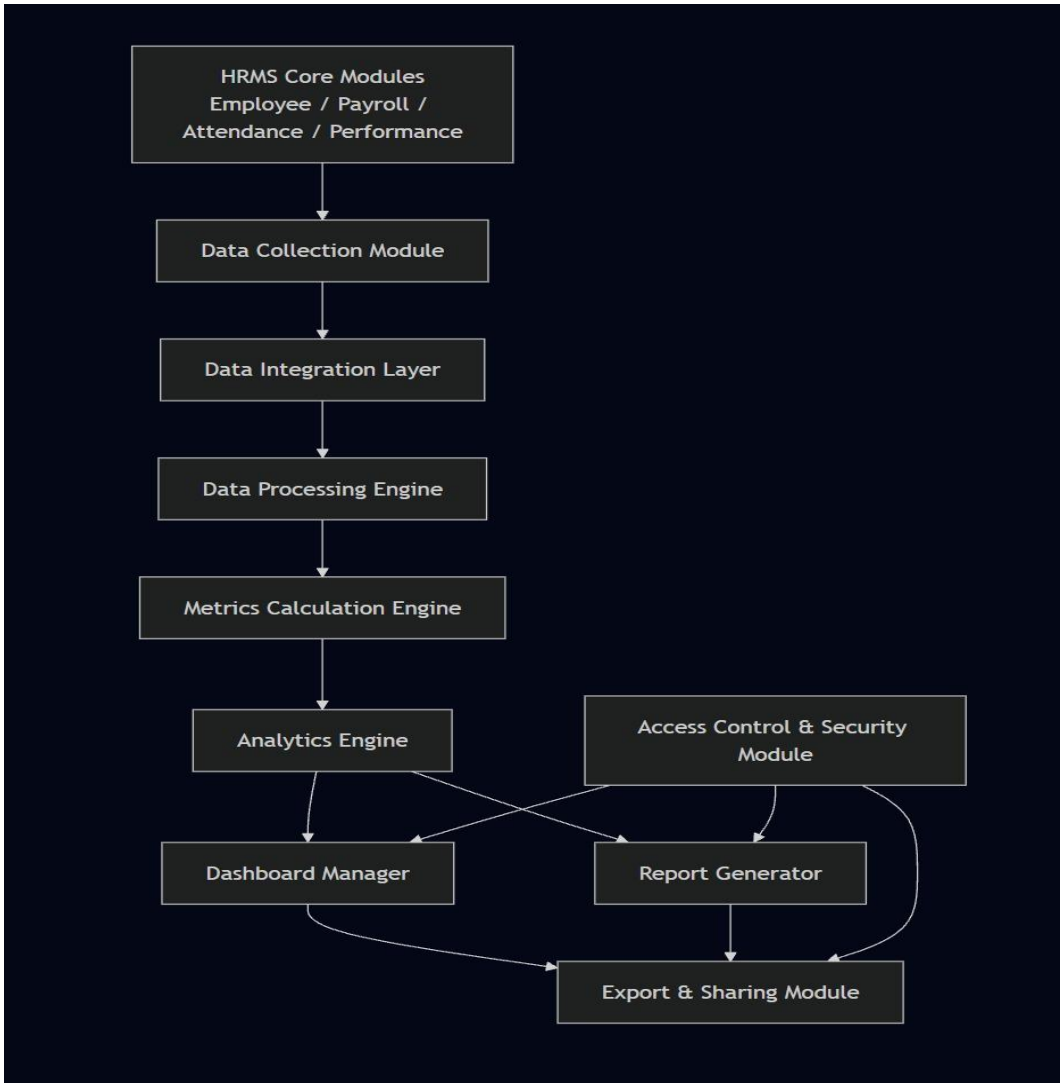
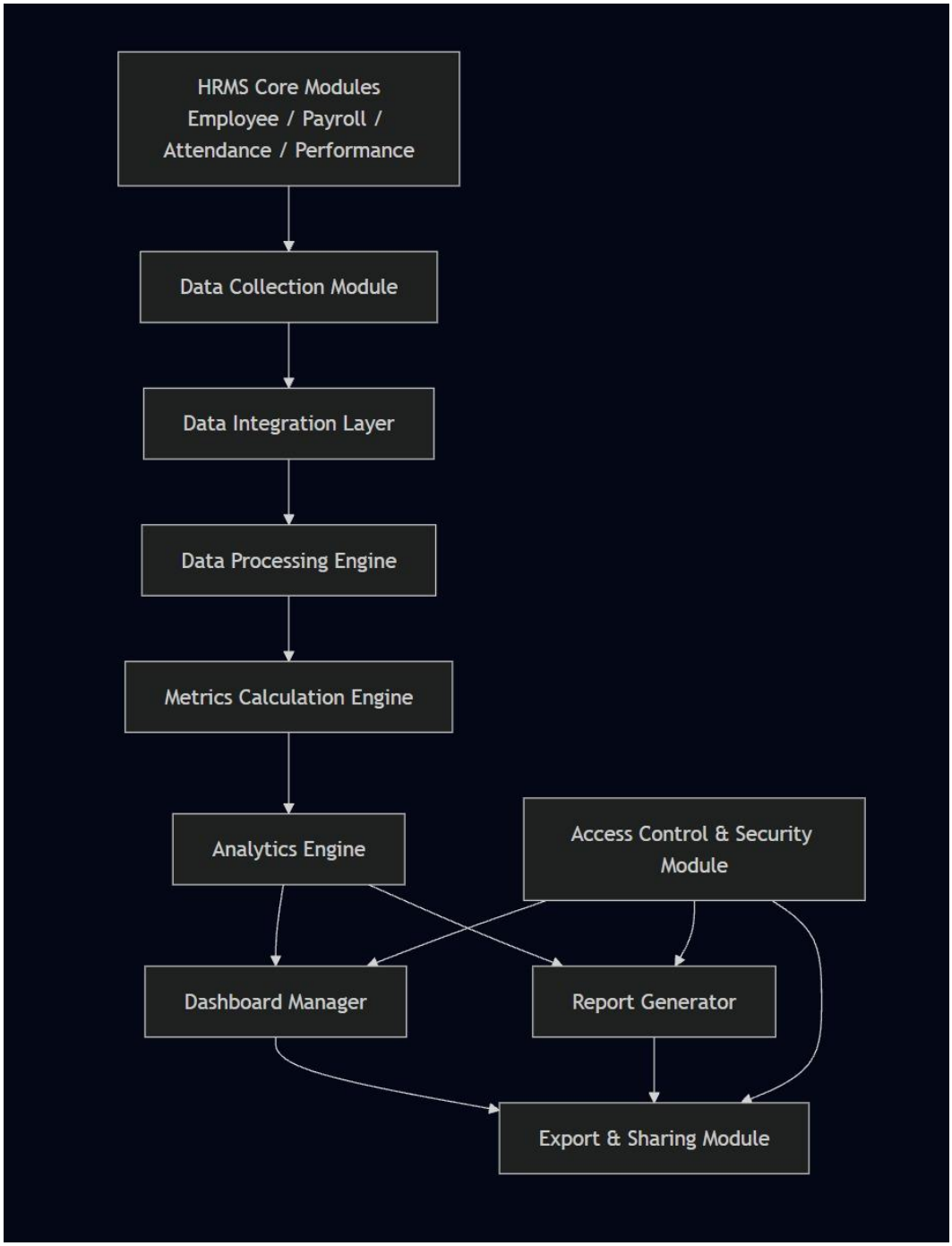
1. Backend Architecture Flowchart

This diagram illustrates the internal data processing pipeline for the HR analytics sub-system. Raw data flows from HRMS Core Modules (Employee, Payroll, Attendance, Performance) into the Data Collection Module. It then passes through the Data Integration Layer, Data Processing Engine, and Metrics Calculation Engine. The Analytics Engine generates insights which are delivered to the Dashboard Manager and Report Generator. The Access Control & Security Module governs authorization and secure access. Finally, the Export & Sharing Module enables report distribution.



2. UI Wireframe Flowchart

This wireframe represents the front-end structure of the HR Analytics Dashboard. The layout begins with a header and filtering options for selecting departments, date ranges, and employee categories. Below this, KPI cards display metrics such as total headcount, attrition rate, and average performance score. The central section contains visual charts such as Employee Growth and Attrition Analysis. At the bottom, action buttons allow users to generate reports and export analytics data.